

Q4) Trace the following code and Show what will appear on the screen shows the following characters at the beginning.

[illegible]

Draw the screen after executing the following code:

```
mov ah, 06h
mov al, 3
mov cx, 0006h
mov dx, 0709h
int 10h
```

[illegible]

INT 10H		
AH	Input Parameters	Description
06	AL: Number of lines to scroll. BH: Screen attribute for cleared area. CL: x coordinate UL, CH: y coordinate UL DL: x coordinate LR, DH: y coordinate LR	Clear or scroll up.

0111, 0101

Q1) Trace the following fragment of code and answer the questions below:

^{SR}
MOV AL, [800H]
MOV CL, 4
ROL AL, CL
MOV [800H], AL

DS	
800H	57
801H	A2
802H	6B
803H	CE
804H	43

- A. What is the value of memory location (DS:800H) = ~~85~~ ²
- B. Modify the previous code to apply the same operation on the 5 memory locations starting at (DS:800H). (Note: use LOOP instruction).

mov cx, 5 ✓
mov si, 800H ⁰⁻⁵
LI: push cx ✓
mov AL, [si] ✓
mov cl, 4
ROL AL, CL
mov [si], AL
pop cx
inc si
loop LI

4
etc

Q2) Consider the following piece of stack memory, suppose that SS = 100h and that the physical addresses increase in the direction of the arrow. Trace the following code and fill the blanks with the correct answer:

PUSH 4A2CH

;SP= 11FEH

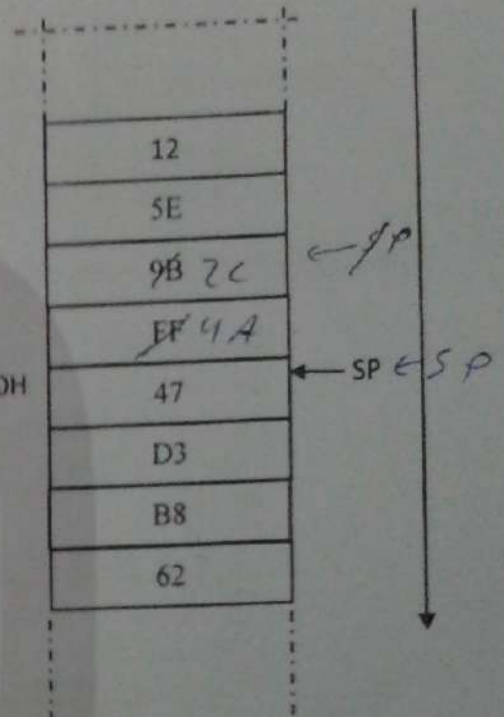
POP AX

;AX= 4A2CH

POP BX

;BX= D398, SP= 1302H

01200H



Q3) Trace the following piece of codes and fill the blanks with the correct answers:

a)
MOV AL, -2
MOV BL, 2
MUL BL

Ax = 1E4H

b)
MOV AX, -5
MOV BL, 2
IDIV BL

Ax = F102H

c)
X DB 'ABC'
MOV SI, 0
MOV CX, 3
L: TEST X[SI], 01H
JZ L1
ADD X[SI], 20H
L1: INC SI
LOOP L

X = 'aBc'

Q2) Consider the following piece of stack memory, suppose that SS = 100h and that the physical addresses increase in the direction of the arrow. Trace the following code and fill the blanks with the correct answer:

PUSH 4A2CH

;SP= 11FEH

POP AX

;AX= 4A2CH

POP BX

;BX= D397, SP= 1203H

01200H

12
5E
9B 2C
4A
47
D3
B8
62

← SP

← SP ← SP

Trace and fill the blanks with the correct answers: